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Surface growth with temporally correlated noise

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We simulate ballistic deposition with long-range temporal correlated noise of bounded amplitude in 1 + 1 dimensions. Good agreement with the dynamical renormalization-group calculation of Medina *et al.* [Phys. Rev. A **39**, 3053 (1989)] is obtained for the scaling exponents when the noise is generated by a version of Mandelbrot's fast fractional Gaussian noise (FFGN) generator. However, using the original FFGN, other exponents are obtained. We suggest that this error is due to an extraordinarily slow crossover caused by the existence of an anomalous growth mode incompatible with the KPZ equation. This may have implications on similar model-dependent results for recent simulations on growth with power-law noise and spatially correlated noise.

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Nonequilibrium surface growth phenomena often exhibit a phenomenon called kinetic roughening, where the surface develops a self-affine morphology [1]. For a system of size L, the surface width $W(L,t)$ at time t can then be expressed in the scaling form $W(L,t) = L^\alpha f(t/L^z)$ where α and z are the roughness and dynamical exponents respectively. The universal scaling function f has specific asymptotic properties such that $W(L,t) \sim t^\beta$ for $t \ll L^z$ and $W(L,t) \sim L^\alpha$ for $t \gg L^z$ where $\beta = \alpha/z$ is the early-time exponent. Much attention has been focused on a special class of models which are described by the KPZ equation [2]:

$$\frac{\partial h(x,t)}{\partial t} = \nu\nabla^2 h(x,t) + \frac{\lambda}{2}(\nabla h(x,t))^2 + \eta(x,t), \tag{1}$$

where $h(x,t)$ and $\eta(x,t)$ are respectively the surface height and the noise at position x and time t. For the $1+1$ dimension, when the noise $\eta(x,t)$ is uncorrelated and has a bounded distribution, both theory and simulations give exponents $\alpha = \frac{1}{2}$ and $\beta = \frac{1}{3}$.

Medina *et al.* [3] investigated theoretically the case when the noise has long-range spatio-temporal correlations and bounded amplitude. The correlator they considered takes the asymptotic form

$$< \eta(x,t)\eta(x',t') > \sim | x - x' |^{2\rho-1} | t - t' |^{2\theta-1} \tag{2}$$

in the $1+1$ dimension. Using a dynamical renormalization-group (DRG) method, they found that for sufficiently small ρ and θ, which correspond to correlations of short range, the critical exponents are the same as for uncorrelated noise. For ρ or θ beyond certain critical values, the correlations become relevant and the surface scaling exponents are nontrivial functions of ρ and θ.

There have been several numerical investigations of surface growth with spatial correlations alone [4]. For example, using a restricted solid-on-solid model with the noise generated by a Fourier transform method, Amar, Lam, and Family found excellent agreement between the simulated values of the surface scaling exponents and the DRG result. However, significant disagreements are reported for some other models [4].

In this paper we investigate another case, namely noise with temporal correlations only ($\rho = 0$), which is a more crucial test of the DRG method. This is because, for the case of spatial correlation, there exist two exponent identities, from which α and β can be obtained exactly as a function of ρ [3]. However, in the presence of temporal correlation, the exponents have to be calculated by the full DRG numerically. Medina *et al.* found that, for $0.167 < \theta < 0.5$, the correlation is relevant and the exponents can be fitted to

$$\alpha(\theta) = 1.69\theta + 0.22\ , \qquad \beta(\theta) = \frac{(1+2\theta)\alpha(\theta)}{2\alpha(\theta)+1}. \tag{3}$$

In this work we check (3) by simulating a ballistic deposition model with temporally correlated noise. More details will be reported elsewhere.

We need to generate random variables $\eta(x,t)$ that satisfy Eq. (2) without spatial correlation. For each fixed lattice site x, $\eta(x,t)$ is an independent fractional Gaussian noise sequence [5] parametrized by t. We generate them using a generalization of Mandelbrot's "fast fractional Gaussian noise (FFGN) generator" [6]. The FFGN X_t is defined by $X_t = \sum_{n=1}^{N} W_n X_t(u_n)$. Each component $X_t(u)$ has autocorrelation decaying at rate u and is constructed by

$$\begin{aligned} X_1(u) &= (1-r^2)^{-1/2}[\xi_1(u) - 0.5], \\ X_t(u) &= rX_{t-1}(u) + [\xi_t(u) - 0.5] \qquad \text{for } t \geq 2, \end{aligned} \tag{4}$$

where $r = e^{-u}$ and the $\xi_t(u)$'s are uncorrelated uniform deviates in the range 0–1. The decay rates are chosen to be $u_n = aB^{-n}$ and the weight function is given by

$$W_n^2 = \frac{12(1-r_n^2)}{\Gamma(2-2\theta)}(B^{1/2-\theta} - B^{-(1/2-\theta)})(aB^{-n})^{1-2\theta}. \quad (5)$$

To generate FFGN with the expected correlation extending up to a time difference T, the number of components needed is $N = \| \log(aQT)/\log(B)\|$ where $\|y\|$ is the smallest integer above y and Q satisfies

$$Q^{2\theta-1} = [-\log(1-\theta_2)]^{1-2\theta}(1-\theta_3) + \theta_3/\theta(2\theta+1). \quad (6)$$

We put $\theta_2 = 0.01$, $\theta_3 = 0.05$, B=2 ,and a=6.

We tested the noise by directly computing the correlator. The correlator follows Eq. (2) nicely for $\theta \lesssim 0.3$, even for time difference $t - t'$ as small as 1. For larger θ, the measured exponent is smaller than expected and is as small as 0.40 for $\theta = 0.45$. In the following, the θ we quote is the measured exponent instead of the nominal input value of the generator. The range of the power-law correlation is also smaller than the nominal value T for large θ. In that case we make sure that the correlator is long range enough by checking that increasing T will not change the ballistic deposition result.

We define a mapped FFGN noise X_t' by

$$X_t' = \begin{cases} 1 & \text{if } X_t > 0 \\ 0 & \text{otherwise.} \end{cases} \quad (7)$$

The quality of the correlation is practically the same as that for the original noise X_t. The motivation for this further mapping will be disussed below.

We simulated ballistic deposition with temporally correlated noise defined by the following evolution equation:

$$h(x,t+1) = \max\{h(x,t)+\eta(x,t),\ h(x-1,t),\ h(x+1,t)\} \quad (8)$$

where $\eta(x,t)$ corresponds to the size of the particle dropped to site x at time t. This size varies randomly in a temporally correlated way and is determined by the mapped FFGN generator described above: $\eta = X'$. The surface height $h(x,t)$ is an integer variable in this case. Periodic boundary conditions and parallel updating is used so that growth is attempted at all even (odd) sites at even (odd) time steps.

Both the roughness exponent α and the early-time exponent β are measured. For α, we simulated deposition onto an initially flat surface with lattice size L varying from 30 to 1200. We associate a surface relaxation time with the average time during which the surface width rises up to half of the steady-state value. The nominal correlation time T is put to be 1000 times the surface relaxation time. We obtained the average saturated surface width $W(L)$ from measurements taken after the surface roughness has fully developed. Especially for simulations with large L and large θ (which implies large z in this case) when the surface relaxation time is long, very long runs of up to about 400 000 time steps and an average over 10 runs were done to obtain reasonable statistics. Log-log plots of $W(L)$ against L give straight lines for the whole range of L and θ investigated. The value of α obtained from the slope of the log-log plot is shown in Fig. 1(a), where the solid line shows the DRG prediction.

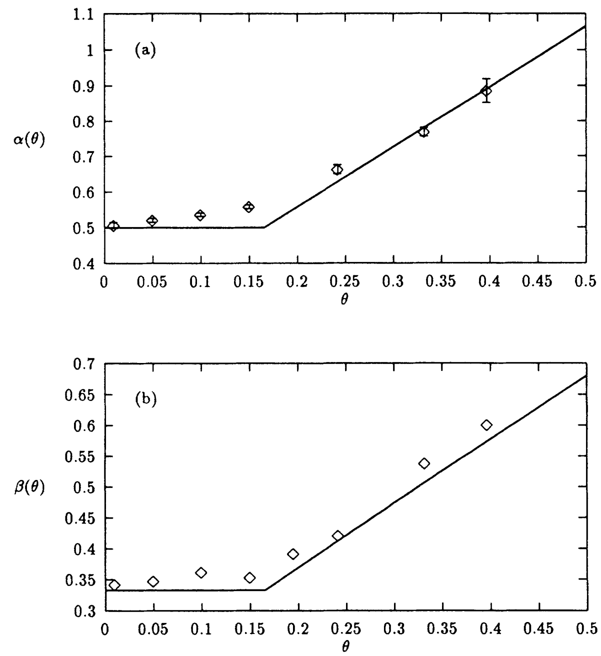

FIG. 1. Ballistic deposition simulation results of (a) surface roughness exponent α and (b) early-time exponent β as functions of θ, which relates to the exponent in the decay of the temporal correlation of the mapped fractional Gaussian noise (FFGN) X_t'. The error bars for α indicate the fitting error only. The solid lines show the dynamical renormalization-group (DRG) results from Ref. [3].

To measure β, we did similar simulations with a large lattice size $L = 32\,760$. A time-dependent surface width $W(t)$ averaged over independent runs was measured well before the saturation of the fluctuations. We found that the "intrinsic width" [7] of the surface is rather large compared to the uncorrelated noise case so that a log-log plot of $W(t)$ against t is linear typically only for $t \gtrsim 50$ for $\theta \leq 0.25$, and even worse for larger values of θ. We simulated growth up to 5000 and 20 000 time steps for θ smaller or greater than 0.25, respectively. An average is taken over 10 and 4 independent runs, respectively. The value of the nominal correlation time T varies from 10^8 up to 10^{11}. The near linear region is fitted to the functional form $W(t) = (t^{2\beta} + W_0^2)^{1/2}$ where the free parameter W_0 corresponds to the intrinsic width. The resulting value of β for various values of θ is shown in Fig. 1(b) together with the DRG result.

We found good agreement between our simulations and the DRG prediction for both α and β. The region of comparatively large discrepancy is centered around the expected critical point at $\theta = 0.167$. The values of the exponents there also deviate from the exact identity $2\rho - 1 - 2\alpha + (2\theta + 1)z = 0$, which holds generally for the correlator in Eq. (2) and results from the fact that the correlator is not renormalized [3]. Therefore the discrepancy is not due to any approximations in the DRG calculation. We suspect that this is a crossover effect in the simulation.

In this investigation, we found that the measured values of the exponents depend considerably on properties of the noise other than the correlator. For θ approaching 0, the noise becomes temporally uncorrelated and the exponents α and β should approach $\frac{1}{2}$ and $\frac{1}{3}$, respectively. This occurs for the mapped FFGN X_t', as already seen in Figs. 1(a) and 1(b). We also simulated deposition using the original FFGN X_t, in which case $h(x,t)$ becomes a continuous variable, and found that the exponents are significantly different. At $\theta = 0.01$, in particular, we got $\alpha = 0.56$ and $\beta = 0.39$. Simulations for larger θ indicate a rise in the value of α but at a much slower rate, so that α becomes smaller than the DRG prediction for large θ. For the $\theta = 0.01$ case, the exponents are consistent with the identity $\alpha + \alpha/\beta = 2$ due to the invariance of Eq. (1) under an infinitesimal tilt [8, 3], which is expected to hold only when there is no temporal correlation. This indicates that the spurious exponents at $\theta \sim 0$ result from reasons other than the temporal correlation. We will suggest an explanation for this apparent inconsistency.

Let us look at the aggregates themselves. Figures 2(a) and 2(b) show snapshots of growing aggregates with the original FFGN X_t and the mapped noise X_t', respectively for $\theta = 0.15$. For the original noise, there are large sparsely occupied regions in the bulk of the aggregate, which are absent for the mapped noise case. We suggest that the sparse regions are responsible for the erroneous result in the scaling exponents.

Consider now deposition with the X_t. The positive long-range correlation in the noise implies that if a large noise value is sampled at a particular time step, it is also likely that we will have large noise values at that site at the subsequent time steps. Thus there is an appreciable possibility that the noise at a particular site maintains a value significantly larger than the average noise magnitude, say the variance σ_η of the noise η, for a long duration. The surface at that point will then be much higher than the neighbors and give rise to large steps. According to the deposition rule in Eq. (8), the neighboring sites will grow by sticking to the side near the top of the large steps. This propagates the steps laterally and leaves overhangs beneath. This is exactly the same as the formation of analogous empty regions observed for ballistic deposition with power-law noise [10]. In the present case, during the period when the noise remains large at that site, new steps are created at each time step and thus an array of laterally propagating fronts lying on top of each other will be formed that leaves an array of overhangs constituting the sparse region. However, for the mapped noise, a large magnitude of noise does not exist simply because the noise is constrained to be 0 or 1. The sparse region is not present, as observed from the snapshot.

The growth of the sparse regions is, in fact, not described by the KPZ equation. This can be checked by examining the lateral surface advance velocity v_l. During the growth of the sparse region, from the evolution Eq. (8), the array of overhangs advance one lattice step laterally per unit time, leading trivially to a constant $v_l = 1$, irrespective of the local slope or curvature of the surface. This contradicts the prediction of the KPZ equation. The growth of the sparse region is "anomalous." From Fig. 2(a), we see that this is not strictly a microscopic effect. The overhangs can typically propagate freely for as far as 20 lattice sites or more. In addition, the effect on the roughness can be large because the surface roughness is affected mostly by the larger noise, which is exactly what characterizes anomalous growth.

We expect that the anomalous growth does not alter the asymptotic exponents, but only contributes to an extraordinarily slow crossover. Recall that large steps on the surface can be created and propagate only when their

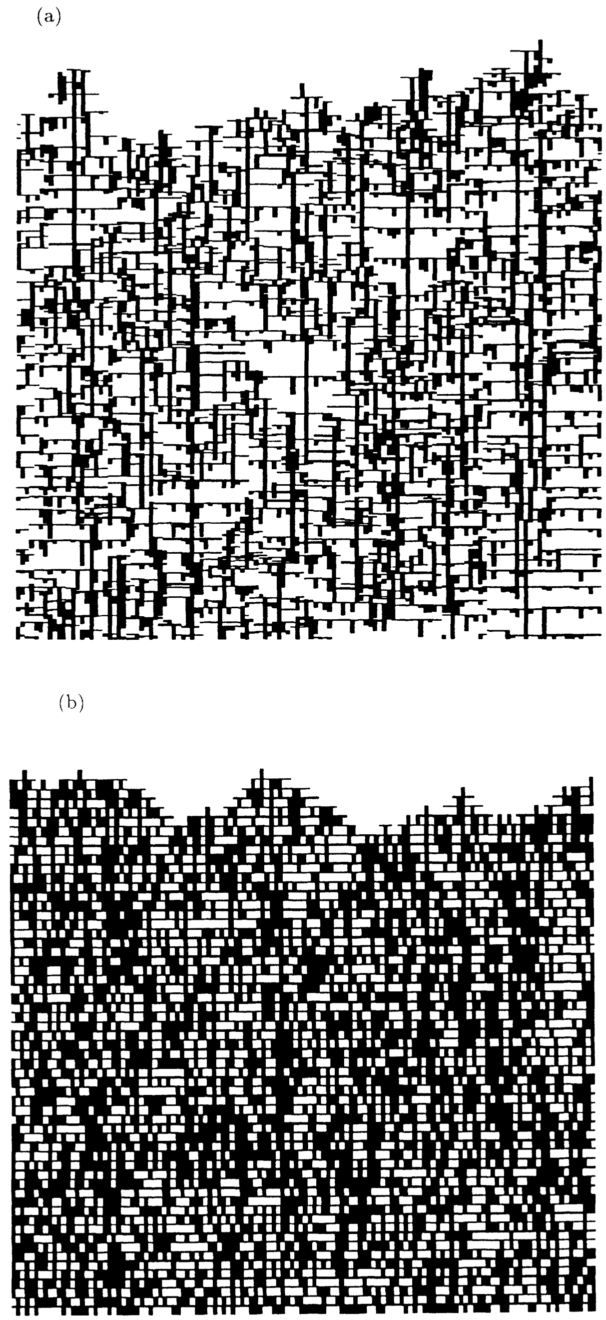

FIG. 2. Snapshots of growing ballistic deposition aggregates with (a) original FFGN X_t and (b) mapped FFGN X_t'. Lattice size $L = 128$ and the exponent of the correlation $\theta = 0.15$. Particles of very small size are represented as of finite small size to show their presence.

height is much larger than σ_η. During anomalous growth when there exists an array of propagating steps, the local slope has to be greater than σ_η. However, for surfaces that we are considering, with roughness exponent $\alpha < 1$, the local slope decreases as we coarse grain the surface. At sufficiently large scales, the coarse-grained slope is always much smaller than σ_η and no array of steps can be present at that level of resolution. Thus the size of the array of overhangs is bounded and anomalous growth should vanish at large scales. In order to see no overhangs in Fig. 2(a), the resolution would have to be decreased by one to two orders of magnitude. To obtain the true asymptotic exponents, we may need a lattice size 10 to 100 times larger than in the case when anomalous growth is suppressed.

Finally, we suggest a qualitative picture to illustrate how anomalous growth can increase the estimated roughness exponent α for $\theta \simeq 0$. During a given period from time t_1 to t_2, the vertical surface displacement $\Delta h = h(x,t_2) - h(x,t_1)$ at any site obeys

$$\Delta h \geq \sum_{t=t_1}^{t_2-1} \eta(x,t). \tag{9}$$

This represents the fact that there can be voids inside, so the surface displacement is generally more than simply stacking the particles together. However, assume that the noise at site x remains large during that period, giving rise to anomalous growth. The first term on the right-hand-side of Eq. (8) for site x is always the maximum of the three and Eq. (9) becomes an equality. This causes the continuous vertical shafts in Fig. 2(a) inside the sparse regions at the site with large noise. No void occurs on that column and enhancement of surface advance by coupling with neighbors through sideways sticking is completely cut off. Since anomalous growth puts the surface advance at the site of the large fluctuation at the lowest possible limit set by Eq. (9), it is not surprising that it reduces the surface roughness compared with KPZ description or simulations where anomalous growth is absent. However, the effect of anomalous growth vanishes at large scale. Therefore the rate of increase of surface roughness with respect to the lattice size is faster with anomalous growth and amounts to a higher measured value of the exponent α.

Other model-dependent results of exponents occur in simulations of surface growth with long-range spatially correlated noise [4] or with power-law noise [9]. For the power-law noise case, the same anomalous growth exists [10] and is much more prominent. The success in the current work of understanding the effect of anomalous growth and properly suppressing them may have implications for these systems [11].

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